# Route optimization of simultaneous delivery and pick-up in automotive inbound logistics under the milk-run mode in a dual-carbon background

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## Abstract

**Purpose** – Under the "dual-carbon" goal, this study optimized the Two-echelon Vehicle Routing Problem with Time Windows and Simultaneous Delivery and Pickup (2E-VRPSPDTW) in automotive inbound logistics under a milk-run mode to minimize the total cost and carbon emissions while maximizing time-window satisfaction.
**Design/methodology/approach** – A novel multi-objective mixed-integer programming model was developed for a three-echelon network comprising suppliers, distribution centers and an original equipment manufacturer. The model incorporates industry-specific constraints, such as two-stage operations, differentiated volume constraints for nested empty and loaded containers and time-window satisfaction requirements. An improved non-dominated sorting genetic algorithm II (NSGA-II) with enhanced operators was designed.
**Findings** – Validation using real third-party logistics provider data shows that the algorithm outperforms benchmarks (MOEA/D, SPEA2 and Pareto-TS) in terms of solution quality, convergence and diversity, particularly for small-to-medium instances. Furthermore, sensitivity analysis confirmed that carbon tax and time-penalty costs significantly influence routing decisions and system performance.
**Originality/value** – This study clarifies the essential differences between the classical VRPSPDTW and the 2E-VRP by treating empty container flow as an active management variable. It incorporates the physical characteristics of nested empty containers and double-carbon constraints, thus providing a targeted modeling framework and an efficient algorithm for the synergistic optimization of economic, environmental and service benefits in automotive inbound logistics.


**Keywords** Automotive inbound logistics, Milk-run, 2E-VRPSPDTW, Dual-carbon background, The improved NSGA-II algorithm

**Paper type** Research article

## 1. Introduction

### *1.1 Background*

Against the global macro backdrop of addressing climate change and advancing a green, low-carbon transition, the strategic goals of "carbon peaking and carbon neutrality" pose transformative demands across all industries (Zhang *et al.*, 2015; Zhou *et al.*, 2018; Sadeq *et al.*, 2024). As a major source of carbon emissions, the transportation sector must undergo a green transformation to support the achievement of dual carbon goals (Wu and Wu, 2022; Ye *et al.*, 2022). The automotive industry involves extensive supply chain systems and complex logistics activities (Zhu and Wu, 2013). Among those, the automotive inbound logistics segment, which transports components from suppliers to the production lines of original equipment manufacturers (OEM), represents a critical yet challenging breakthrough point for carbon reduction in the automotive sector owing to its high-frequency, multi-node operations and energy-intensive nature. Optimizing inbound logistics operations and transportation routes has significant practical implications for reducing the overall carbon footprint of the supply chain and enhancing resource efficiency (Che and Yang, 2023; Zhou and Wen, 2023).

*Competing interests:* The author declares no conflict of interest with any financial organizations regarding the material reported in this manuscript.

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Among various inbound logistics models, the milk-run system has been widely adopted in the automotive manufacturing industry because of its consolidated and highly efficient characteristics (Bocewicz *et al.*, 2021). In this model, a third-party logistics (3PL) provider coordinates and dispatches vehicles to follow optimized, fixed routes, sequentially visiting multiple suppliers in close geographical proximity (Wang and Chen, 2019). During these trips, vehicles simultaneously pick up standardized loaded containers with components required for production and return standardized empty containers, thereby achieving a closed-loop integration of forward material flow and reverse container flow. These “simultaneous pickup and delivery” operations effectively consolidate scattered transportation demands and reduce empty vehicle runs and certain loading/unloading activities, demonstrating significant advantages over traditional direct shipping in terms of lowering transportation costs and improving container turnover rates (Tang *et al.*, 2020).

*1.2 Problem statement*

Among the vehicle routing problems used to represent automotive inbound logistics, the Vehicle Routing Problem with Time Windows and Simultaneous Delivery and Pickup (VRPSPDTW) and the Two-echelon Vehicle Routing Problem (2E-VRP) are the two most commonly utilized. The VRPSPDTW is typically defined as a single-layer network problem in which vehicles depart from a distribution center to simultaneously deliver goods to customer nodes and pick up returned items. These pickups and deliveries are subject to customer time window constraints. Conversely, the 2E-VRP typically describes a transportation network comprising “factory–transit warehouse–customer” nodes and its main focus is the consolidation and redistribution of goods at intermediate points. Although the constructed Two-echelon Vehicle Routing Problem with Time Windows and Simultaneous Delivery and Pickup (2E-VRPSPDTW) model resembles the VRPSPDTW and 2E-VRP, it exhibits fundamental distinctions in several key aspects (Figure 1).

First, goods are limited to modular parts that can be directly integrated into the four major processes of the OEM – stamping, welding, painting and assembly – explicitly excluding procurement logistics that require further processing (e.g. components or raw materials needed by module manufacturers). Suppliers interface directly with the production processes of the

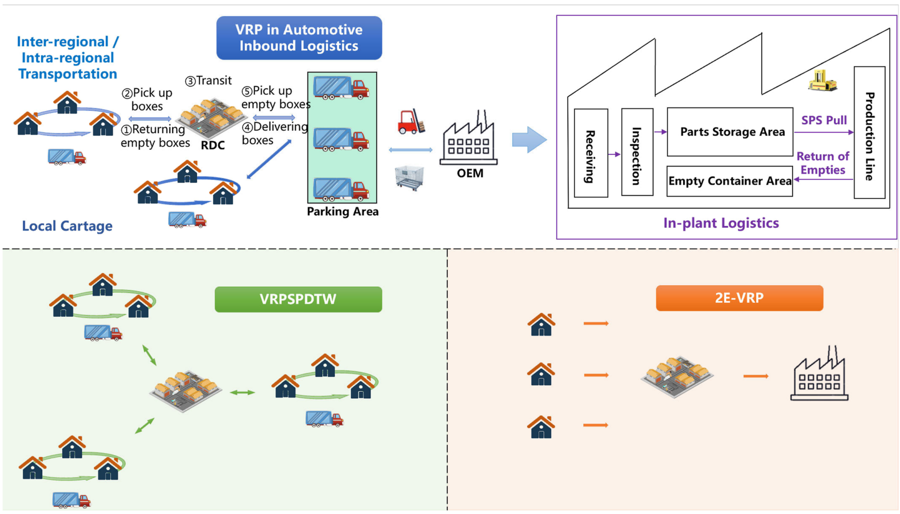


**Figure 1.** Schematic of the operational processes

OEM and thus serve both as the source of full containers and recipients of empty containers. Second, the model is tailored to a third-party logistics-led operational mode, where the distribution center acts as a core hub and undertakes functions such as container consolidation, loading/unloading operations, temporary storage and empty container scheduling, thereby bridging the milk-run and loading-delivery phases. Third, it focuses on the transportation characteristics of "multiple suppliers, small batches, and high frequency" under the milk-run model, in which the reverse flow of empty containers is coordinated alongside the forward delivery of full containers. This characteristic demonstrates distinct industry-specific traits that align with the logistics patterns of the automotive parts supply for leveled production in OEMs.

The specific differences between VRPSPDTW and the proposed model in this study are as follows:

(1) Network Hierarchy and Process Phased. The proposed model explicitly divides the transportation process into two distinct stages: the "milk-run" and the "loading and delivery" phases. In the milk-run phase, vehicles visit suppliers solely to perform "pickup of full containers and return of empty containers." In the loading and delivery phases, vehicles depart from the distribution center to deliver full containers to the OEM and pick up empty containers. Vehicles are prohibited from serving across stages. This phase separation reflects the scheduling logic of material consolidation and just-in-time delivery in automotive manufacturing. Conversely, in VRPSPDTW, vehicles perform mixed pickup and delivery tasks within a single trip.

(2) Differentiated Constraints for Empty and Full Containers. In automotive logistics, empty standard foldable containers can be nested to occupy approximately one-fifth of the volume of full containers, with negligible weight. Consequently, modeling requires setting separate volume constraints for full and empty containers rather than treating them under a simple unified load capacity. This physical characteristic is rarely considered in general VRPSPDTW formulations and represents a unique packing and loading constraint specific to automotive part logistics.

(3) Transitivity of Time Windows and Measurement of Satisfaction. The time windows in this study are not directly imposed on the OEM but are applied to the processing time of goods arriving at the distribution center. These windows subsequently influence the timeliness of the final delivery to the OEM through the operating schedule of the distribution center. Time satisfaction is measured by the proportion of goods delivered to the distribution center within a specified time window. This implication reflects the operational characteristic of the "distribution center acting as a time buffer and coordination node" in automotive inbound logistics.

The specific differences between 2E-VRP and the proposed model in this study are as follows:

(1) Bidirectional Flow and Operations. The 2E-VRP involves unidirectional logistics, whereas the proposed model incorporates bidirectional logistics interactions at each level. Between suppliers and the distribution center, full containers are picked up and empty containers are returned; between the distribution center and the OEM, full containers are picked up and empty contained are collected. This bidirectional flow originates from the operational model of standardized container reuse in automotive manufacturing, which represents a typical closed-loop logistics system.

(2) Container Management as a Core Constraint. This paper explicitly incorporates the supply and recovery of empty containers as part of scheduling decisions, assuming that the distribution center possesses sufficient empty container resources to support the production cycle of suppliers. Such a coordinated allocation of container resources is generally not included in traditional 2E-VRP models and represents a unique

constraint in the context of standardized material packaging and a green circular economy in automotive inbound logistics.

In summary, existing models fall short in addressing the problems specific to the automotive industry. Classical VRPSPDTW models are insufficient for depicting multilayer networks and phased operations, whereas 2E-VRP focuses on the hierarchical integration of unidirectional logistics and lacks descriptions of bidirectional container flows and simultaneous delivery and pickup operations. Therefore, constructing a novel routing optimization model that accurately reflects the practical features of milk-run operations in the automotive industry and integrates multidimensional optimization objectives has significant theoretical value and practical urgency.

The model proposed in this paper constitutes a vehicle routing problem that integrates a two-echelon network structure, a two-phase operational process, differentiated constraints for empty and full containers, time-window transitivity satisfaction evaluation and carbon emission cost considerations. It is neither a simple extension of the VRPSPDTW to a multilayer network nor a traditional 2E-VRP augmented with pickup and delivery operations. Rather, it is a novel optimization problem with industry specificity and contemporary relevance, which was developed based on the context of milk-run operations in automotive manufacturing, incorporating the operational characteristics of third-party logistics and the constraints imposed by dual-carbon policies.

### *1.3 Objectives*

The objectives of this model are to minimize transportation costs and carbon emissions and maximize the service level. This study explores the problem of automobile inbound logistics in the milk-run pick-up mode from multiple perspectives. It aims to provide adequate decision-making information for route planning for simultaneous delivery and pickup in automotive inbound logistics and to analyze the impact of current low-carbon policies on enterprises. This helps to scientifically plan vehicle routes and realize a balance between corporate interests and environmental benefits.

The remainder of this paper is organized as follows. Section 2 reviews and analyzes the evolution of research on the VRPSPDTW and multi-objective optimization algorithms. Section 3 develops a mixed-integer mathematical programming model that reflects the 2E-VRPSPDTW specificity in automotive inbound logistics. Section 4 proposes an INSGA-II to address the 2E-VRPSPDTW. Section 5 validates the effectiveness and robustness of the proposed improved NSGA-II (INSGA2) algorithm through real-world case studies, comparative experiments and sensitivity analyses. Section 6 derives targeted policy recommendations and managerial implications based on the research findings. Finally, Section 7 summarizes the findings and provides an outlook for future research.

## 2. Literature review

The optimization of automotive inbound logistics, particularly under the milk-run mode, constitutes a critical operational challenge that directly aligns with the strategic “dual-carbon” goals of reducing carbon emissions and fostering sustainable development. This complex operational task is fundamentally categorized as a VRPSPDTW involving the coordinated collection of components and the return of empty containers within strict production windows (Yu and Chen, 2011).

The inherent NP-hard complexity of VRPSPDTW, as demonstrated by computationally intensive exact algorithms, such as the branch-and-price algorithm, has focused research on heuristic and metaheuristic solutions (Angelelli and Mansini, 2002). For instance, Mingyong and Erbao (2010) proposed an improved differential evolution algorithm that incorporates penalty functions and enhanced operators to mitigate premature convergence. Ru (2024) applied a Tabu search algorithm to optimize intermodal routes in vehicle logistics to increase profits for enterprises. Some studies have explored hybrid strategies to enhance solution efficiency and quality. Wang and Mu (2014) designed a parallel simulated annealing algorithm, while Mu *et al.* (2015) integrated a residual capacity and radial surcharge heuristic with an improved simulated annealing process. Furthermore, a collaborative evolutionary genetic algorithm (Wang and Chen, 2012) and a Memetic algorithm designed for related split-delivery problems (Liu *et al.*, 2021) have enriched the methodological framework. Hof and Schneider (2019) combined path relinking with an adaptive large neighborhood search to significantly improve the convergence speed and solution quality. Wu and Gao (2023) introduced directional random transition rules and disruption operators into ant colony optimization to enhance search capability and avoid local optima. García-Nájera *et al.* (2015) proposed a similarity-based selection multi-objective evolutionary algorithm for vehicle routing problems with backhauls. Lv and Shen (2023) developed an improved NSGA-II algorithm that incorporates a local search to simultaneously optimize inventory costs, tardiness and makespan. These routing principles have been applied to the specific context of automotive inbound logistics, often integrating the milk-run strategy. Chen *et al.* (2021) established an integrated model combining milk-run collection with other strategies, optimizing it via an MILP model and genetic algorithm to reduce mileage and energy consumption. Baller *et al.* (2022) applied an MILP approach in a real-world case study to achieve substantial reductions in both cost and $CO_2$ emissions, thereby directly linking operational optimization with environmental objectives.

Under the “dual-carbon” imperative, the research focus has evolved from predominantly cost-centric, single-objective models toward comprehensive multi-objective optimization frameworks. Amongst the most significant work is that of Mahmoodi and colleagues in developing multi-objective models for complex routing problems (Mahmoodi *et al.*, 2022a, b). Their work includes a robust optimization model for delivery networks that integrates cost, time and risk (Mahmoodi *et al.*, 2022a, b), and an enhanced NSGA-II framework to minimize cost, service time, risk and battery consumption under the consideration of $CO_2$ emissions (Mahmoodi *et al.*, 2024). Furthermore, they explored advanced hybrid algorithms by combining NSGA-II with Bayesian belief networks for UAV location routing (Mahmoodi and Hashemi, 2025; Mahmoodi *et al.*, 2025). Similarly, Abbasi *et al.* (2024) designed a green logistics network under carbon tax policies using a fuzzy inference system and multiobjective programming. Zhang *et al.* (2025) developed a multistage hybrid evolutionary algorithm (MS-HEMO-MRSS) for a multi-vehicle-type VRPSPDTW that demonstrated superior performance. These studies underscore the feasibility of integrating economic, environmental and risk-based objectives. Furthermore, research such as that of Frey *et al.* (2023), who solved a problem with capacitated flexible delivery locations using a hybrid adaptive large neighborhood search, highlights methods for incorporating greater operational realism.

Table 1 presents an overview of the relevant literature. Despite these advancements, a critical gap remains in the specific domain of automotive inbound logistics. Current VRPSPDTW research remains largely limited to single-stage networks, with insufficient attention being paid to the two-echelon structure (multiple suppliers, distribution centers and a single OEM) intrinsic to this field. Moreover, while multi-objective optimization is gaining traction, its application to this three-echelon milk-run problem under explicit “dual-carbon” constraints – such as distinct loading constraints for stacked empty containers and loaded units – is under-explored. Therefore, this study aims to bridge this gap by developing a multi-objective optimization model for the 2E-VRPSPDTW problem in automotive milk-run logistics. The objective is to develop a model which simultaneously minimizes the total logistics cost and carbon emissions, incorporates practical separate-volume constraints and is solved using an improved metaheuristic algorithm.

## 3. 2E-VRPSPDTW mathematical model

This study examines the 2E-VRPSPDTW algorithm in automotive inbound logistics during the milk-run pickup and loading transportation phases (Figure 2). In the milk-run pick-up phase, vehicles are dispatched from the distribution center to specific suppliers at the OEM to pick up loaded components while returning empty logistics containers to the suppliers. In the loading and delivery phase, vehicles dispatched from the distribution center carry out a

**Table 1.** Overview of relevant literature

| References | Two-echelon | Milk-run | Simultaneous delivery and pick-up | Time windows constraints | Empty container stacking constraints | Multi-objective | Algorithm | Datasets |
|---|---|---|---|---|---|---|---|---|
| Wang and Chen (2012) | | | √ | √ | | | Metaheuristic Algorithm | Benchmark |
| Wang and Mu (2014) | | | √ | √ | | | Heuristic Algorithm | Benchmark |
| García-Nájera *et al.* (2015) | | | √ | | | √ | Metaheuristic Algorithm | Benchmark |
| Mu *et al.* (2015) | | | | | | √ | Heuristic Algorithm | Benchmark |
| Hof and Schneider (2019) | | | √ | √ | | | Metaheuristic Algorithm | Benchmark |
| Chen *et al.* (2021) | √ | √ | | | | | Metaheuristic Algorithm | Case |
| Liu *et al.* (2021) | | | √ | √ | | | Metaheuristic Algorithm | Benchmark and Case |
| Baller *et al.* (2022) | | √ | | | | | Exact Algorithm | Case |
| Zhou *et al.* (2022) | √ | | | √ | | | Metaheuristic Algorithm | Benchmark |
| Mahmoodi *et al.* (2022a, b) | | | | | | √ | Metaheuristic Algorithm | Case |
| Frey *et al.* (2023) | | | | √ | | | Metaheuristic Algorithm | Benchmark and Case |
| Lv and Shen (2023) | | | | | | √ | Metaheuristic Algorithm | Benchmark |
| Wu and Gao (2023) | | | √ | √ | | | Metaheuristic Algorithm | Benchmark |
| Abbasi *et al.* (2024) | | | | | | √ | Exact Algorithm | Case |
| Ru (2024) | √ | | | √ | | | Metaheuristic Algorithm | Case |
| Mahmoodi *et al.* (2025) | | | √ | √ | | √ | Metaheuristic Algorithm | Case |
| Zhang *et al.* (2025) | | | √ | √ | | √ | Metaheuristic Algorithm | Benchmark |
| This work | √ | √ | √ | √ | √ | √ | Metaheuristic Algorithm | Case |

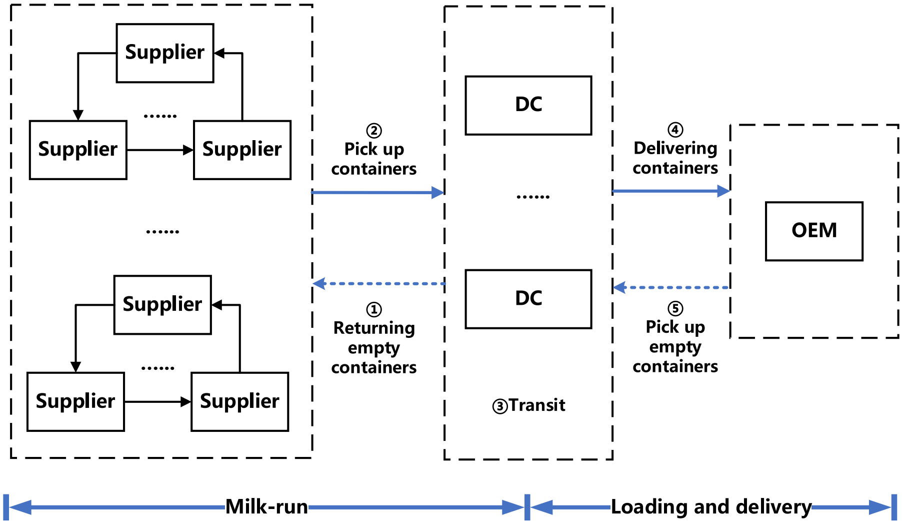


**Figure 2.** Automobile inbound logistics

centralized distribution of components from different suppliers to the OEM within a specified time, whereas empty containers are retrieved from the OEM. This section describes the problem and the proposed mixed-integer programming model.

### *3.1 Notations*

*3.1.1 Sets and indices.* The 2E-VRPSPDTW problem is defined as follows: Let $G = (V, A)$ be a directed graph with the set $V = M \cup S_i \cup R_j$ of nodes, where $M$ represents the OEM; $S_i$ denotes the set of auto parts suppliers, $i = 1, 2, \ldots, I$ and $I$ represents the number of auto parts suppliers. $R_j$ denotes the set of distribution centers, $j = 1, 2, \ldots, J$, $J$ represents the number of distribution centers. $A = \{(p, l) : p, l \in V, p \neq l\}$ denotes the set of edges. $F_{wk}$ represents the vehicle collection, $w$ represents the model number, $w = 1, 2, \ldots, W$, $W$ represents the number of vehicle models. $k$ is the vehicle number of the $w$ th model, $k = 1, 2, \ldots, K$ and $K$ represents the number of vehicles of the $w$ th model.

*3.1.2 Parameters.* The parameters involved in the proposed mixed-integer programming model are as follows: The distance between nodes $p$ and $l$ is denoted as $d$. Specifically, $d_{ji}$ is the distance from the distribution center $j$ to supplier $S_i$; $d_{ij}$ is the distance from supplier $S_i$ to distribution center $j$. $d_{ie}$ is the distance from supplier $S_i$ to the next supplier $S_e$. $d_{jm}$ is the distance from the distribution center $j$ to the OEM. $d_{mj}$ is the distance from the OEM to distribution center $j$. $q_i$ is the total cargo weight required by the OEM from supplier $S_i$; $Q_{wk}$ is the rated load capacity of the $F_{wk}$ vehicle type. $N_i$ is the number of loaded containers provided by suppliers. $N_m$ is the number of containers required by the OEM from all suppliers, $N_{wk}$ is the maximum number of cases that can be placed in a vehicle. $EN_i$ indicates the number of containers required by the supplier, $EN_j$ denotes the number of empty containers owned by distribution center $j$ and $EN_m$ indicates the number of empty containers required by the OEM for recycling. $EN_{wk}$ is the maximum number of empty containers that can be placed in a vehicle. $c_{wko}$ denotes the fixed cost of dispatching vehicle $F_{wk}$; $c_{wkt}$ denotes the unit transport cost of vehicle $F_{wk}$; $c_{ja}$ denotes the unit dispensing and handling cost of the distribution center; $c_{js}$ denotes the unit storage cost of the distribution center; $Wh_j$ denotes the maximum number of cargo containers stored in the distribution center; and $v_j$ denotes the speed of cargo handling in the distribution center.

$v_{twk}$ is the transportation speed (km/h) of vehicle $F_{wk}$; $v_{swk}$ is the loading and unloading speed of vehicle $F_{wk}$; $T_j$ indicates the total processing time for cargo to reach the distribution center $j$; $E_1 T_{ai}$

indicates the earliest time specified for the supplier $S_i$ to reach the distribution center $j$; $E_2T_{ai}$ indicates the earliest acceptable time for the supplier $S_i$ to reach the distribution center $j$; $L_1T_{ai}$ indicates the latest time specified for the supplier $S_i$ to reach the distribution center; $L_2T_{ai}$ indicates the latest acceptable time for supplier $S_i$ to reach the distribution center $j$; $r = T_{iawk}$ indicates the time point at which vehicle $F_{wk}$ actually transports the cargo from the supplier $S_i$ to the distribution center; $T_{ojwk}$ indicates the time point at which vehicle $F_{wk}$ departs from the distribution center $j$ and is set up at $T_{ojwk} = 0$; $T_{ijwk}$ indicates the length of transport required for vehicle $F_{wk}$ from supplier $S_i$ to distribution center $j$, $T_{jiwk} = T_{ijwk}$; $T_{iewk}$ indicates the length of transport required for the vehicle from supplier $S_i$ to supplier $S_e$; $T_{iwk}$ indicates the actual length of stay of the vehicle $F_{wk}$ at the supplier $S_i$; $\mu_1$ indicates the early delivery penalty factor; $\mu_2$ indicates the late delivery penalty factor; $E_a^g$ indicates the physical quantity of fuel consumed at the end-use stage; $\gamma$ is the standard coal reference factor; $\tau$ indicates the fuel conversion to standard coal emission factor; and $f_c^w$ indicates the fuel consumption per kilometer for the $w$ th vehicle type.

*3.1.3 Decision variables.* The decision variables are defined as follows. $A_{iwk}$ indicates the actual number of empty containers returned by a single vehicle during the circulation phase; $EA_{iwk}$ indicates the actual number of containers picked up by a single vehicle during the circulation phase; $A_{mwk}$ indicates the actual number of containers transported to the main plant by a single vehicle during the distribution phase; $EA_{iwk}$ indicates the actual number of empty containers recovered from the OEM by a single vehicle during the distribution phase; and $N_j$ is the quantity of loaded containers transported from the supplier to the distribution center $j$. $y_{iwk}$ denotes whether vehicle $F_{wk}$ completed the pickup and empty-container return service for supplier $S_i$ during the milk-run phase (completed service = 1, incomplete service = 0). $y_{ijmwk}$ denotes whether vehicle $F_{wk}$ completed cargo transportation from the distribution center $j$ to the OEM for supplier $S_i$ during the loading and delivery phases (complete transport = 1, incomplete transport = 0). $y_{jwk}$ denotes whether vehicle $F_{wk}$ belongs to distribution center $j$ (yes = 1, no = 0). $z_{ijwk}$ represents whether vehicle $F_{wk}$ traveled directly from supplier $S_i$ to distribution center $j$ (direct travel = 1, indirect travel = 0). $z_{iewk}$ denotes the direct vehicle travel from supplier $S_i$ to the next supplier $S_e$ (direct travel = 1, indirect travel = 0). $z_{mjwk}$ represents the direct vehicle travel from the OEM to distribution center $j$ (direct travel = 1, indirect travel = 0). $z_{imwk}$ represents direct vehicle travel from supplier $S_i$ to the OEM (direct travel = 1, indirect travel = 0). $z_{juwk}$ represents the direct vehicle travel from distribution center $j$ to the next distribution center, (direct travel = 1, indirect travel = 0). $\varphi_i$ represents cargo delivery from supplier $S_i$ within the specified time window $[E_1T_{ai}, L_1T_{ai}]$ (yes = 1, no = 0).

### *3.2 Model assumptions*

(1) All suppliers, distribution centers and OEM locations are known in advance.

(2) The OEM contains information about the required quantity of cargo from each supplier and the number of empty containers that are picked up. The number of empty containers picked up from each supplier is also known.

(3) The distribution centers have an unlimited supply of empty containers and can meet the OEM's demands.

(4) The spatial constraints at the distribution centers do not consider empty containers; only the capacity constraints on the loaded containers are considered.

(5) The weight of stacked empty containers is significantly lower than that of the vehicle payload. Therefore, the payload constraints do not consider the weight of empty containers.

(6) All cargo is transported using standard reusable containers with dimensions 1200 mm × 1000 mm × 1000 mm. Containers are stackable; therefore, an empty container occupies one-fifth of the space occupied by a loaded container.

(7) Each distribution center in the system has constraints on the number of vehicles and available space. Each distribution center can collect cargo from multiple suppliers and deliver it to the OEM.

(8) Each vehicle belongs exclusively to one distribution center and can either participate in transportation or remain idle. If a vehicle is involved in transportation, it can only be engaged in either the milk-run or the loading and delivery phases, but not both. Furthermore, vehicles must return to their original distribution centers after completing their tasks.

(9) Suppliers cannot transport cargo directly to the OEM; horizontal transportation between distribution centers is prohibited.

(10) After the milk-run phase, the cargo destined for the distribution centers must undergo centralized processing at the respective distribution centers before proceeding to the loading and transportation phase.

(11) During the milk-run phase, the demand for both loaded and empty containers from suppliers is assigned to a single distribution center and handled by a single vehicle. Partial pick-up or partial return of empty containers is not allowed.

(12) During the loading and delivery phase, the distribution center that picks up the loaded containers is responsible for delivering them to the OEM. Only one vehicle from the distribution center is assigned to transport the containers, which cannot be delivered in partial shipments. However, multiple vehicles in separate batches can fulfill the OEM's request for an empty container pickup.

(13) The time window for the OEM's actual acceptance of supplier cargo is simplified to the time it takes to reach the distribution center and undergo processing.

*3.3 Objective functions*

Considering the profitability of the enterprise, the demand for automobile inbound logistics to maintain the OEM's operation and the social responsibility to respond to the "dual-carbon" policy, objective functions are constructed based on three dimensions: cost, time satisfaction and carbon emissions.

*3.3.1 Formulating the cost objective.* Objective Function 1 (cost minimization) is given by Eq. (1).

$$MinZ = C_1 + C_2 + C_3 + C_4 \tag{1}$$

where $C_1$ represents the cost of vehicle activation, as shown in Eq. (2).

$$C_1 = \sum_{w=1}^{W} \sum_{k=1}^{K} c_{wko} y_{wk}^{n} y_{jwk} x_j \tag{2}$$

$C_2$ represents the transportation cost, as shown in Eq. (3).

$$C_2 = \sum_{w=1}^{W} \sum_{k=1}^{K} \sum_{j=1}^{J} c_{wkt} \left( d_{ji} y_{wk} y_{jwk} x_j + d_{ij} y_{wk} y_{jwk} x_j + d_{jm} y_{wk} y_{jwk} x_j + d_{mj} y_{wk} y_{jwk} x_j + d_{ie} y_{iwk} \right) \tag{3}$$

$C_3$ represents the distribution center consolidation cost, as shown in Eq. (4).

$$C_3 = \sum_{j=1}^{J}\sum_{w=1}^{W}\sum_{k=1}^{K}\sum_{i=1}^{I} c_{ja}N_i y_{iwk} y_{wk} y_{jwk} x_j + \sum_{j=1}^{J} x_j c_{js} T_j \tag{4}$$

$C_4$ represents the penalty cost, as shown in Eq. (5).

$$C_4 = f(r) = \begin{cases} \mu_1 * q_i(E_2T_{ai} - r) & , E_2T_{\mathrm{ai}} \le r < E_1T_{\mathrm{ai}} \\ 0 & , E_1T_{\mathrm{ai}} \le r \le L_1T_{\mathrm{ai}} \\ \mu_2 * q_i(r - L_2T_{ai}) & , L_1T_{\mathrm{ai}} < r \le L_2T_{\mathrm{ai}} \\ \infty & , \quad \text{others} \end{cases}, r = T_{iawk} \tag{5}$$

*3.3.2 Formulating the carbon emissions objective.* Objective Function 2 (carbon emission minimization) is given by Eq. (6).

$$MinE^g_{co_2} = E^g_a \gamma \tau \tag{6}$$

$E^g_a$ represents the physical fuel consumption of the fuel terminal as expressed in Eq. (7).

$$E^g_a = \sum_{w=1}^{W}\sum_{k=1}^{K}\sum_{j=1}^{J} f^w_c \left(d_{ji} y_{wk} y_{jwk} x_j + d_{ij} y_{wk} y_{jwk} x_j + d_{jm} y_{wk} y_{jwk} x_j + d_{mj} y_{wk} y_{jwk} x_j + d_{ie} y_{iwk}\right) \tag{7}$$

*3.3.3 Formulating the service level objective.* Objective Function 3 (service level maximization) is expressed in Eq. (8).

$$Max\eta = \alpha\eta_1 + \beta\eta_2 \tag{8}$$

where $\alpha$ and $\beta$ are weighting coefficients satisfying $\alpha + \beta = 1$.

$\eta_1$ represents the temporal satisfaction rate, defined as a continuous function based on time deviation, as shown in Eq. (9).

$$\eta_1 = \frac{\sum_{i=1}^{I} N_i f_i(t_i)}{\sum_{i=1}^{I} N_i} \times 100\% \tag{9}$$

Here, $f_i(t_i)$ denotes the satisfaction function for an order defined in Eq. (10).

$$f_i(t_i) = \begin{cases} \dfrac{t_i - E_2T_{ai}}{E_1T_{ai} - E_2T_{ai}} & , E_2T_{\mathrm{ai}} \le t_i < E_1T_{\mathrm{ai}} \\ 1 & , E_1T_{\mathrm{ai}} \le t_i \le L_1T_{\mathrm{ai}} \\ \dfrac{L_2T_{ai} - t_i}{L_2T_{ai} - L_1T_{ai}} & , L_1T_{\mathrm{ai}} < t_i \le L_2T_{\mathrm{ai}} \\ 0 & , \quad \text{others} \end{cases} \tag{10}$$

$\eta_2$ is the order fulfillment rate, as defined in Eq. (11) and is measured as the percentage of cargo delivered within a specified time.

$$\eta_2 = \frac{\sum_{i=1}^{I} S_i \varphi_i}{\sum_{i=1}^{I} S_i} \times 100\% \tag{11}$$

*3.4 Model constraints*

This section describes the various constraints within the constructed multi-objective mixed-integer programming model. The core objective of the model constraints is to align the routing optimization scheme with the practical operational scenario of automotive inbound logistics to ensure feasibility, compliance and a balance between efficiency and constraint boundaries, thereby achieving a multi-objective equilibrium among cost, carbon emissions and temporal satisfaction. The practical logic and operational mechanisms of the constraints across multiple dimensions are explained below.

*3.4.1 Transportation route constraints.* This set of constraints primarily governs the routing logic for vehicles during the milk-run and loading and delivery phases, which form a closed-loop workflow, to ensure the integrity and consistency of the transportation process. Route constraints are designed to regulate vehicle flow, clarify accountability and prevent disorganized transportation. For example, requiring vehicles to return to their original distribution centers facilitates scheduling management and subsequent maintenance. Direct shipments from suppliers to the OEM and transfers between distribution centers are prohibited because the milk-run model mandates consolidated processing; goods are first gathered at the distribution center for sorting and loading before being delivered collectively to the OEM. This constraint avoids increased costs and carbon emissions associated with fragmented transportation.

3.4.1.1 Vehicle return-to-depot constraints. During the milk-run and loading and delivery phases, vehicles must depart from their assigned distribution center $j$ and return to the same distribution center upon the completion of their transportation tasks. This constraint reflects the practical operational requirements for clear vehicle assignments and manageable dispatching in logistics operations. Eq. (12) and Eq. (13) describe the milk-run stage and the loading and delivery stage constraints, respectively.

$$\sum_{i=1}^{I} z_{jiwk} = \sum_{i=1}^{I} z_{ijwk} \tag{12}$$

$$z_{jmwk} = z_{mjwk} \tag{13}$$

3.4.1.2 Supplier service uniqueness constraint. During the milk-run stage, only one vehicle from a single distribution center is responsible for the pick-up and return of empty containers at supplier $S_i$, as shown in Eq. (14). This constraint simplifies the delineation of transportation responsibility and facilitates the organization and tracking of loading and unloading operations.

$$\sum_{w=1}^{W} \sum_{k=1}^{K} y_{iwk} y_{jwk} = 1 \tag{14}$$

3.4.1.3 Vehicle phase mutual exclusivity constraint. $F_{wk}$ vehicles can either participate in the milk-run or loading and delivery stages but cannot simultaneously participate in both stages. This constriction is expressed in Eq. (15) and reflects the specialization of vehicles in task assignment, which is conducive to improving the operational efficiency in each phase.

$$z_{jiwk} + z_{jmwk} \leq 1 \tag{15}$$

3.4.1.4 Transportation continuity constraint. This constraint ensures the continuity of transportation between the milk-run and loading and delivery stages, as expressed in Eq. (16) and Eq. (17).

$$\sum_{i=1}^{I} z_{jiwk} = \sum_{i=1}^{I} z_{ijwk} \tag{16}$$

$$z_{jmwk} = z_{mjwk} \tag{17}$$

3.4.1.5 Route prohibition constraints. Direct shipments from suppliers to the OEM and direct transfers of goods between different distribution centers are prohibited. This is a typical setup in most outsourced inbound logistics models, which requires all goods to be consolidated, sorted and loaded at the distribution center to leverage economies of scale and process control advantages. Supplier $S_i$ cannot directly deliver to the OEM, as expressed in Eq. (18). A distribution center cannot directly deliver to another distribution center $j$, as expressed in Eq. (19).

$$\sum_{w=1}^{W} \sum_{k=1}^{K} z_{imwk} = 0 \tag{18}$$

$$\sum_{w=1}^{W} \sum_{k=1}^{K} z_{juwk} = 0 \tag{19}$$

3.4.1.6 Vehicle dispatch activation constraints. Vehicles must be dispatched during the milk run and the loading and delivery stages at the distribution center to ensure that transportation demands are met and service disruptions caused by vehicle idleness are prevented. This constraint is described using Eq. (20) and Eq. (21).

$$\sum_{i=1}^{I} \sum_{w=1}^{W} \sum_{k=1}^{K} z_{jiwk} \geq x_j \tag{20}$$

$$\sum_{w=1}^{W} \sum_{k=1}^{K} z_{jmwk} \geq x_j \tag{21}$$

*3.4.2 Transport volume constraints.* This set of constraints primarily governs the flow volumes of goods and empty containers to ensure a supply–demand balance and compliance with vehicle loading capacities. Resources, such as vehicle weight or volume limits and the storage capacity of distribution centers, are finite. Furthermore, the volumetric difference between full and empty containers affects the loading efficiency, thus rendering it essential to impose constraints to prevent overloading, stockouts or site congestion.

3.4.2.1 Goods transportation consistency constraint. Goods picked up from suppliers must be delivered to the OEM by the same distribution center during the loading and delivery phases; split deliveries are not permitted. This constraint ensures clear transportation accountability and the integrity of goods traceability. In the milk-run stage, the cargo of supplier $S_i$ is picked up by a vehicle from a specific distribution center. In the loading and delivery stage, only one vehicle from the distribution center is responsible for transporting cargo to the OEM. Cargo cannot be delivered in batches. These constraints are expressed using Eq. (22).

$$\sum_{w=1}^{W} \sum_{k=1}^{K} y_{ijmwk} y_{jwk} = 1 \tag{22}$$

3.4.2.2 Flow conservation constraints. The quantity of cargo arriving at the distribution center during the milk-run stage is equal to the quantity transported from the distribution center to the manufacturing plant during the loading and delivery stage. This constraint is described using Eq. (23). The total quantity of cargo provided by all suppliers is equal to the quantity transported to the distribution center and is equal to that required by the OEM, as described in Eq. (24). This is a fundamental requirement for flow balance in logistics systems.

$$\sum_{w=1}^{W}\sum_{k=1}^{K}\sum_{i=1}^{I}N_i y_{iwk} y_{jwk} = N_j = \sum_{w=1}^{W}\sum_{k=1}^{K}\sum_{i=1}^{I}N_i y_{ijmwk} z_{jiwk} \tag{23}$$

$$\sum_{i=1}^{I}N_i = \sum_{i=1}^{I}N_j = N_m \left(EN_j = \lambda, \lambda \text{ is a vast positive integer}\right) \tag{24}$$

3.4.2.3 Vehicle loading capacity constraints. This constraint limits the number of full and empty containers loaded per vehicle during both the milk-run and loading and delivery phases to ensure that it does not exceed the maximum volume capacity of the vehicle while also complying with the weight limits, thereby guaranteeing transportation safety and compliance. The quantity of cargo carried by each vehicle does not exceed the maximum capacity of a vehicle. These constraints can be explicitly described using Eq. (25), Eq. (26), Eq. (27) and Eq. (28).

$$A_{iwk} \leq N_{wk} \tag{25}$$

$$A_{mwk} \leq N_{wk} \tag{26}$$

$$EA_{iwk} \leq EN_{wk} \tag{27}$$

$$EA_{mwk} \leq EN_{wk} \tag{28}$$

3.4.2.4 Vehicle activation and transport volume correlation constraints. The actual transportation capacity is generated only when the vehicles are activated, as described in Eq. (29) and Eq. (30).

$$A_{iwk} = \sum_{i=1}^{I}N_i y_{iwk} y_{wk} \tag{29}$$

$$A_{mwk} = \sum_{i=1}^{I}N_i y_{ijmwk} \tag{30}$$

3.4.2.5 Transportation demand fulfillment constraints. The total volume of goods transported by all activated vehicles in their respective phases must satisfy the total demand for that phase to ensure the complete execution of transportation tasks. Working vehicles meet transportation requirements during the milk-run and loading and delivery stages. This constraint can be described using Eq. (31), Eq. (32), Eq. (33) and Eq. (34).

$$\sum_{w=1}^{W}\sum_{K=1}^{K}A_{iwk} y_{wk} = \sum_{i=1}^{I}N_i \tag{31}$$

$$\sum_{w=1}^{W}\sum_{K=1}^{K}EA_{iwk} y_{wk} = \sum_{i=1}^{I}EN_i \tag{32}$$

$$\sum_{w=1}^{W}\sum_{k=1}^{K}A_{mwk}y_{wk}=N_m \tag{33}$$

$$\sum_{w=1}^{W}\sum_{k=1}^{K}EA_{mwk}y_{wk}=EN_m \tag{34}$$

3.4.2.6 Weight constraints. The total weight of the cargo transported by the vehicles should not exceed their load capacity, which represents a fundamental restriction for safe vehicle operation. This constraint is expressed in Eq. (35) and Eq. (36).

$$\sum_{i=1}^{I}q_i y_{wk} y_{iwk}\le Q_{wk} \tag{35}$$

$$\sum_{i=1}^{I}q_i z_{jmwk}\le Q_{wk} \tag{36}$$

*3.4.3 Distribution center-related constraints.* This set of constraints reflects the limitations imposed on the transportation plan by the resources and capacities of the distribution centers. The resources of distribution centers, such as vehicle fleet sizes and storage capacities, are fixed. The purpose of these constraints is to prevent any single distribution center from excessively occupying resources to ensure balanced resource utilization across the entire network.

3.4.3.1 Vehicle quantity limitation. The actual number of $w$ th vehicle types at distribution center $j$ does not exceed its actual quantity, reflecting the finiteness of resources, as described in Eq. (37).

$$\sum_{k=1}^{K}y_{wk}y_{jwk}\le\sum_{k=1}^{K}y_{jwk} \tag{37}$$

3.4.3.2 Inventory capacity constraint. The number of cargos transported to the distribution center by vehicles does not exceed the maximum storage capacity of its storage area, as described in Eq. (38).

$$\sum_{w=1}^{W}\sum_{k=1}^{K}\sum_{i=1}^{I}N_i y_{iwk}y_{wk}y_{jwk}\le Wh_j \tag{38}$$

3.4.3.3 Total vehicle balance. The sum of vehicles owned by each distribution center is equal to the total number of vehicles, thus ensuring complete resource allocation, as described in Eq. (39).

$$\sum_{j=1}^{J}\sum_{w=1}^{W}\sum_{k=1}^{K}y_{jwk}=\sum_{w=1}^{W}\sum_{k=1}^{K}F_{wk} \tag{39}$$

*3.4.4 Time-related constraints.* This set of constraints incorporates time windows, processing times and transportation durations to ensure the temporal feasibility of the transportation plan. The OEM operates under a lean production mode, which demands high sensitivity to the arrival times of the components. Components must arrive exactly on time to sustain production, neither early to avoid inventory build-up nor late to prevent production line stoppage. The essence of these time constraints is to ensure service timeliness and reduce penalty costs.

3.4.4.1 Time window compliance constraint. Ensuring that the cargo from suppliers is delivered within the acceptable time window is crucial, as any delays would result in significant penalty costs and are not considered in this paper. This constraint can be explicitly described using Eq. (40) and reflects the logistics timeliness requirements of the OEM.

$$T_{iawk} = \left(T_{ojwk} + T_{jiwk} + T_{iewk} + T_{iwk} + T_{ijwk} + T_j\right)y_{wk}y_{iwk}y_{jwk} \tag{40}$$

3.4.4.2 Distribution center processing time. Goods undergo loading and consolidation processes after they arrive at the distribution center. The processing duration is proportional to the quantity of the goods. The duration for handling and loading the cargo at the distribution center upon arrival is $T_j$, as described in Eq. (41).

$$T_j = \frac{\sum_{w=1}^{W}\sum_{k=1}^{K}\sum_{i=1}^{I} N_i y_{iwk} y_{wk} y_{jwk}}{v_j} \tag{41}$$

3.4.4.3 Transportation and loading/unloading times. These constraints specify the travel time between nodes and the loading/unloading time at supplier locations, which forms the basis for route time calculations. The duration of transportation from the distribution center to supplier is $T_{ijwk}$, as described in Eq. (42). The duration of a vehicle's transportation between suppliers is $T_{iewk}$, as expressed in Eq. (43). The duration of loading and unloading operations at the supplier's location for a vehicle is $T_{iwk}$, as described in Eq. (44).

$$T_{jiwk} = T_{ijwk} = \frac{d_{ji}}{v_{twk}} \tag{42}$$

$$T_{iewk} = \frac{d_{ie}}{v_{twk}} \tag{43}$$

$$T_{iwk} = \frac{(EN_i + N_i)y_{iwk}y_{wk}}{v_{swk}} \tag{44}$$

3.4.4.4 Vehicle departure time baseline. A departure time of the vehicle from its assigned distribution center is taken to be 0, as described in Eq. (45).

$$T_{ojwk} = 0 \tag{45}$$

3.4.4.5 Transportation time symmetry. The travel times between the distribution center and suppliers, as well as between the distribution center and the OEM, are assumed to be identical for inbound and outbound trips. This assumption simplifies the associated time calculations. The transportation duration of vehicle $F_{wk}$ from the assigned distribution center $j$ to supplier $S_i$ is equal to the duration from supplier $S_i$ to distribution center $j$ and the transportation duration of vehicle $F_{wk}$ from the assigned distribution center $j$ to the OEM is equal to the duration from the OEM to the distribution center. These constraints can be described using Eq. (46) and Eq. (47).

$$T_{jiwk} = T_{ijwk} \tag{46}$$

$$T_{jmwk} = T_{mjwk} \tag{47}$$

*3.4.5 Variables constraints*. This set of constraints defines the value ranges and types of the decision variables, ensuring the mathematical soundness of the model. Decision variables in the model, such as "whether to use a vehicle" or "whether to take a specific route," are central to the solution. The purpose of these constraints is to ensure clearly defined boundaries for these decisions.

Variable constraints are shown in Eq. (48) and Eq. (49).

$$y_{iwk}, y_{ijmwk}, y_{wk}, y_{jwk}, z_{ijwk}, z_{jiwk}, z_{iewk}, z_{mjwk}, z_{jmwk}, z_{imwk}, z_{juwk}, \varphi_i \in \{0, 1\} \tag{48}$$

$$A_{iwk}, EA_{iwk}, A_{mwk}, EA_{iwk}, E_a^g \geq 0 \tag{49}$$

The following section describes the design of the INSGA-II algorithm. This algorithm is used to solve the proposed model, which captures the simultaneous delivery–pickup characteristics described in this section and accounts for cost, carbon emissions and time satisfaction.

## 4. Improved NSGA-II algorithm

The NSGA-II algorithm is commonly used to solve Pareto frontiers (Millar *et al.*, 2023; Mahmoodi *et al.*, 2025). This study designs an enhanced NSGA-II algorithm by improving the crowding distance calculation, elite preservation strategy, crossover probabilities and operators. The design of the proposed enhanced NSGA-II algorithm follows the steps shown in Figure 3.

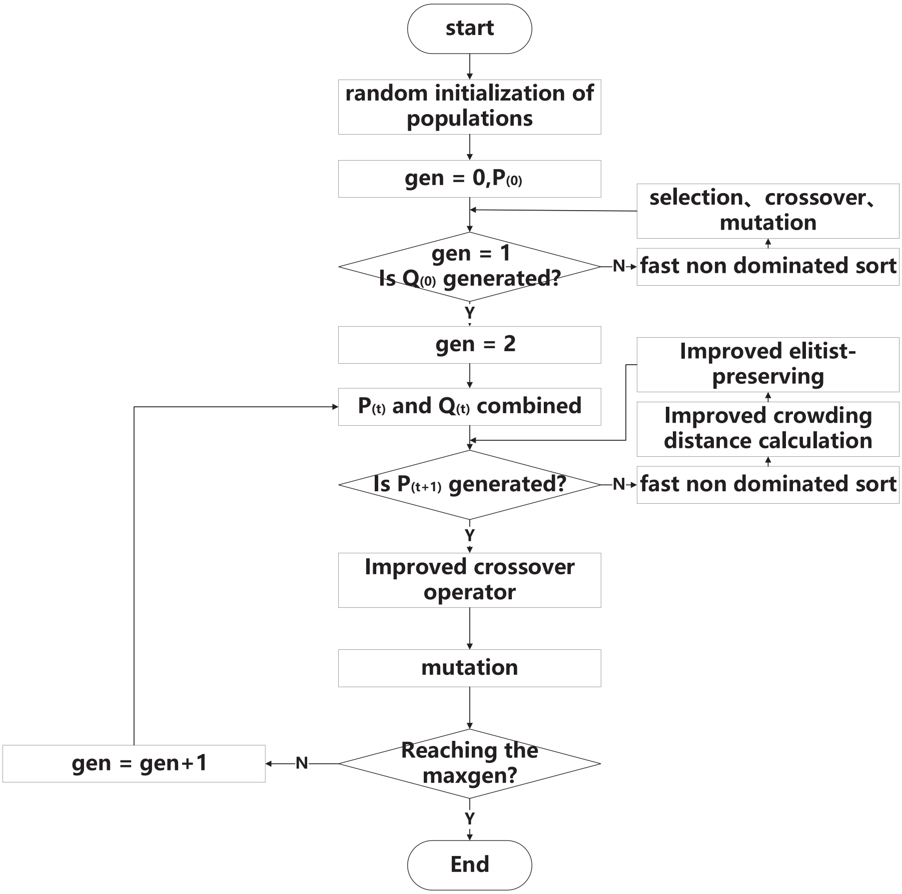


**Figure 3.** INSGA-II steps

*4.1 Improved congestion calculation*

The congestion calculation orders all the solutions into the same class. The NSGA-II uses the congestion degree instead of the fitness sharing strategy, in which the NSGA needs to specify the sharing radius. It serves as the criterion for selecting excellent individuals among peers and ensures diversity within the population. This approach facilitates the selection, crossover and mutation of individuals across the entire range of the objective space, thereby promoting compelling exploration and maintenance of diversity.

To improve the uniformity of individual distribution, the article adopted a dynamic congestion distance evaluation method based on the calculation of congestion distance. The formula is shown in Eq. (47) and Eq. (48).

$$d_{\psi i}^{B} = \frac{d_{\psi i}}{lg\left(\frac{1}{D_i}\right)} \tag{50}$$

$$D_i = \frac{1}{n}\sum_{m=1}^{n}\left(\left|f_m^{i-1} - f_m^{i+1}\right| - d_{\psi i}\right)^2 \tag{51}$$

$D_i$ represents the differences between two adjacent individuals, and $f_k^{i+1}$ and $f_k^{i-1}$ represent the objective function values for individuals $i+1$ and $i-1$ on the kth sub-objective after objective sorting. $f_m^{max}$ and $f_m^{min}$ are the maximum and minimum fitness values for the kth sub-objective after sorting the objectives for individual j, respectively.

*4.2 Improving crossover operation*

Crossover probabilities play a central role in genetic algorithms in determining the probability of two chromosomes making a crossover. Although a high crossover probability enhances the algorithm's ability to open new search regions, it is also associated with an increase in the likelihood of corruption. Low crossover probabilities, conversely, can hinder a search. Therefore, this study uses the adaptive crossover probability to select various probabilities by calculating the chromosome fitness value, as expressed in Eq. (49) and Eq. (50).

$$P_{ac} = mean(P_{aci}) \tag{52}$$

$$P_{aci} = \frac{f_i - min f_i}{max f_i - min f_i} \tag{53}$$

Here, $P_{ac}$ denotes the average value of $P_{aci}$ normalized to the fitness component of each objective, $f_i$ denotes the fitness value of the ith objective function in the population, $max f_i$ denotes the maximum fitness value of the ith objective function in the population and $min f_i$ denotes the maximum fitness value of the ith objective function in the population.

The mechanism of the traditional SBX (simulated binary crossover) operator, which simulates the binary crossover operator, performs crossover operations on real-valued parent individuals. The operator randomly selects a crossover point and exchanges gene codes on both sides of the crossover point. However, because this operator has relatively weak global search capability, an arithmetic crossover operator was used in this study to improve the crossover operation in the NSGA-II algorithm; this enhances its global search capability and maintains population diversity. Let $X_i^t$ and $X_j^t$ be the real-valued encodings of the decision variables for two individuals at the crossover point in a generation. The decision variables for the two individuals after the crossover, denoted as $X_i^{t+1}$ and $X_j^{t+1}$, are calculated using Eq. (51).

$$\begin{cases} X_i^{t+1} = aX_i^t + (1-a)X_j^t \\ X_j^{t+1} = bX_j^t + (1-b)X_i^t \end{cases} \tag{54}$$

where $a$ and $b$ are uniformly distributed random numbers between 0 and 1.

*4.3 Improved elite control strategy*

The elite preservation strategy used in the NSGA-II algorithm yields a rapid reproduction of individuals in the Pareto-optimal solution layers. This reduces the number of non-dominated layers and individuals in other non-optimal dominated layers, thus causing a significant loss of lateral diversity and leading to local convergence. The steps in the original elite preservation strategy are illustrated in Figure 4 (Deb and Goel, 2001).

The elite preservation strategy of the NSGA-II algorithm is changed to the elite control strategy, which is improved such that the maximum number of individuals in each nondominated layer can be constrained to control the number of Pareto-optimal solution layers. Each nondominated layer can be included in the new population to avoid premature convergence. The control elements are shown in Figure 5.

First, the population is sorted based on non-dominance. Assuming that there are 2N individuals in the population, the maximum number of individuals allowed on the jth front (j = 1, 2, . . ., K) in a new population of size N is given by Eq. (52).

$$n_j = N\frac{1-r}{1-r^K}r^{j-1} \tag{55}$$

where $n_j$ is the maximum number of individuals in the jth non-dominant front, $N$ is the population size and $r$ is the reduction rate, $r \in [0, 1)$. K is the number of non-dominant fronts.

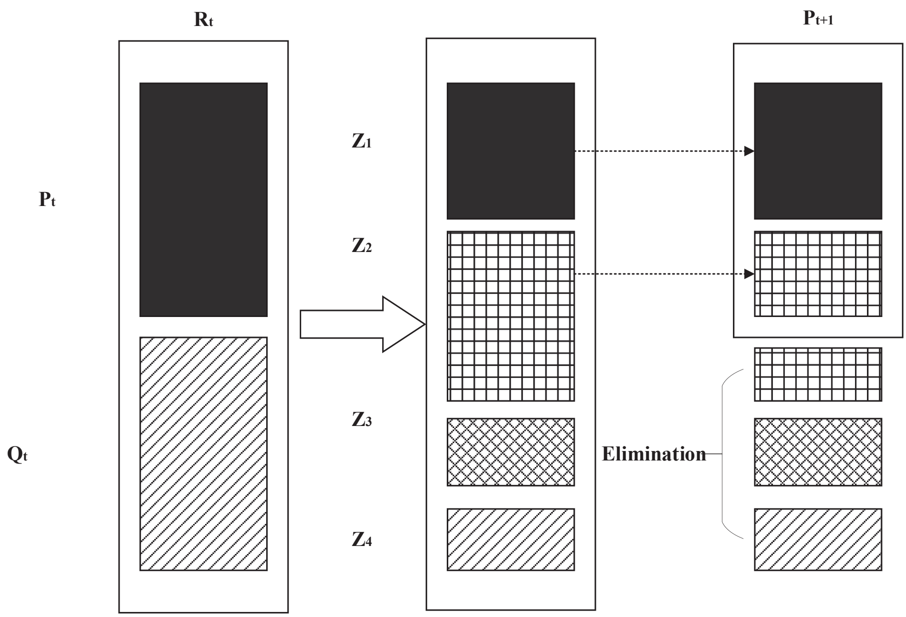


**Figure 4.** Elite preservation strategy execution steps

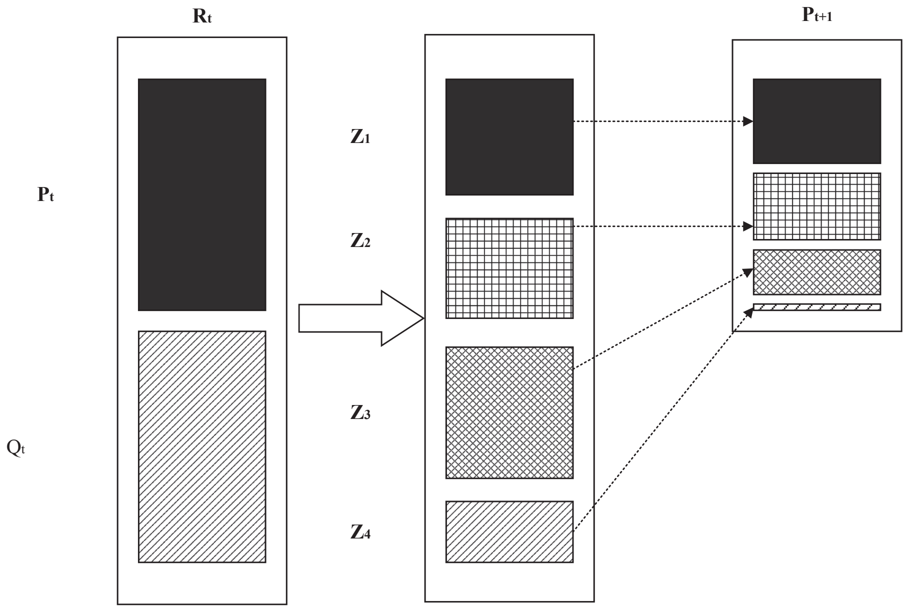


**Figure 5.** Elite control strategy execution steps

Since $r < 1$, the first non-dominant front contains the largest maximum number of individuals. Subsequently, each nondominant front enables an exponentially decreasing number of solutions. The improved elite control strategy allows each nondominant front to be included in the new population, thereby eliminating local convergence.

## 5. Practical applications

### *5.1 Problem background and data description*

The context of this study is based on the actual operations of the third-party automotive inbound logistics company X, which serves the OEM (hereafter referred to as M) in a specific province in China. Company X has two self-owned distribution centers in the province, each equipped with three types of vehicles: 6.8 m, 9.6 m and 13 m. The vehicles handle inbound logistics operations of various OEMs within the province. M outsources its inbound logistics within the province to automotive logistics company X. M has 30 cooperating suppliers, each of which provides different automotive part modules. Combining these thirty-part suppliers, the two distribution centers of automotive logistics companies X and M form a multi-to-one two-echelon network. M follows lean production principles and treats the cargo in transit as its inventory. Therefore, automotive logistics company X has a high frequency of deliveries and must pick up cargo from suppliers and deliver it to the plant daily to maintain production operations.

The inbound logistics distribution business of a specific day was used as a case study to validate the feasibility of the designed VRPSPDTW solution. On a given day, two distribution centers dispatch vehicles to collect cargo from 30 suppliers. After consolidation, the cargo is transported to the OEM. This study provides a reasonable vehicle routing plan for the distribution centers, with the aim of minimizing carbon emissions to support green logistics initiatives, reducing company costs and maintaining a certain level of time satisfaction to enhance competitiveness. Table 2 details information on the 30 suppliers, Table 3 and Table 4

**Table 2.** Node information

| No | X | Y | Number of containers | Number of empty containers | Specified time windows | Actual acceptable time windows | Total weight of cargo(t) |
|---|---|---|---|---|---|---|---|
| M | 31.7 | 407.3 | Total for all suppliers | 650 | – | – | – |
| R1 | 95.9 | 223.1 | – | – | – | – | – |
| R2 | 207.8 | 295.5 | – | – | – | – | – |
| S1 | 192.3 | 229.4 | 3 | 5 | 6:20–8:50 | 5:20–12:20 | 0.969 |
| S2 | 103.7 | 221.1 | 9 | 15 | 5:20–7:50 | 4:20–11:20 | 1.548 |
| S3 | 95.6 | 222.9 | 11 | 20 | 5:20–7:50 | 4:20–11:20 | 4.18 |
| S4 | 104.4 | 215.4 | 15 | 10 | 5:20–7:50 | 4:20–11:20 | 6.405 |
| S5 | 146.9 | 284.6 | 8 | 15 | 6:20–8:50 | 5:20–12:20 | 2.096 |
| S6 | 98.1 | 210.8 | 4 | 5 | 5:20–7:50 | 4:20–11:20 | 1.24 |
| S7 | 48.9 | 178.8 | 13 | 10 | 5:20–7:50 | 4:20–11:20 | 5.343 |
| S8 | 8.9 | 126.2 | 11 | 30 | 5:20–7:50 | 4:20–11:20 | 3.531 |
| S9 | 178.8 | 240.8 | 12 | 15 | 6:20–8:50 | 5:20–12:20 | 5.112 |
| S10 | 225.3 | 313.5 | 15 | 30 | 9:50–12:20 | 8:20–15:20 | 4.725 |
| S11 | 210.6 | 193.2 | 7 | 10 | 9:50–12:20 | 8:20–15:20 | 2.541 |
| S12 | 207.4 | 345.1 | 11 | 10 | 9:50–12:20 | 8:20–15:20 | 3.245 |
| S13 | 161.2 | 139.4 | 5 | 5 | 8:50–11:20 | 7:20–14:20 | 1.665 |
| S14 | 201.2 | 107.6 | 19 | 25 | 9:50–12:20 | 8:20–15:20 | 7.543 |
| S15 | 152.6 | 47.4 | 14 | 20 | 8:50–11:20 | 7:20–14:20 | 3.878 |
| S16 | 176.5 | 86.4 | 50 | 65 | 9:50–12:20 | 8:20–15:20 | 23.5 |
| S17 | 111.3 | 118.7 | 42 | 45 | 8:50–11:20 | 7:20–14:20 | 17.64 |
| S18 | 104.6 | 22.2 | 29 | 30 | 8:50–11:20 | 7:20–14:20 | 10.353 |
| S19 | 112.6 | 24.2 | 27 | 30 | 8:50–11:20 | 7:20–14:20 | 4.293 |
| S20 | 111.9 | 21.7 | 7 | 5 | 9:50–12:20 | 8:20–15:20 | 2.408 |
| S21 | 11.1 | 19.6 | 8 | 10 | 5:20–7:50 | 4:20–11:20 | 1.56 |
| S22 | 42.8 | 43.9 | 10 | 15 | 5:20–7:50 | 4:20–11:20 | 2.73 |
| S23 | 113.2 | 24.1 | 14 | 20 | 9:50–12:20 | 8:20–15:20 | 5.026 |
| S24 | 115.4 | 20.7 | 18 | 15 | 8:50–11:20 | 7:20–14:20 | 6.804 |
| S25 | 111.3 | 24.7 | 6 | 5 | 8:50–11:20 | 7:20–14:20 | 1.932 |
| S26 | 85.6 | 63.4 | 11 | 10 | 5:20–7:50 | 4:20–11:20 | 4.851 |
| S27 | 110.8 | 21.8 | 3 | 5 | 8:50–11:20 | 7:20–14:20 | 0.867 |
| S28 | 108.4 | 19.7 | 9 | 15 | 9:50–12:20 | 8:20–15:20 | 3.051 |
| S29 | 106.1 | 18.8 | 32 | 40 | 8:50–11:20 | 7:20–14:20 | 10.656 |
| S30 | 431.7 | 343.7 | 21 | 30 | 9:50–12:20 | 8:20–15:20 | 9.198 |

**Table 3.** Vehicle information

| | 6.8 m | 9.6 m | 13 m |
|---|---|---|---|
| $Q_{wk}$ | 7t | 16t | 30t |
| $N_{wk}$ | 31 containers | 45 containers | 62 containers |
| $EN_{wk}$ | 124 empty containers | 180 empty containers | 248 empty containers |
| $v_{twk}$ (Km/h) | 65 | 65 | 65 |
| $v_{swk}$ (Container/h) | 120 | 120 | 120 |
| $c_{wko}$ (RMB) | 50 | 60 | 70 |
| $c_{wkt}$ (RMB/km) | 4 | 5.8 | 7.5 |
| $f_c^w$ (RMB/km) | 0.2 | 0.27 | 0.35 |

**Table 4.** Details of the distribution centers

| | R1 | R2 |
|---|---|---|
| $Wh_j$ (Container) | 500 | 375 |
| $v_j$ (Container/h) | 200 | 180 |
| $c_{ja}$ (RMB/container) | 2.50 | 2.40 |
| $c_{js}$ (RMB/containers/h) | 0.33 | 0.26 |

provides specifics of the vehicle information logistics company X and the distribution centers, respectively, and Table 5 lists the values of the other fixed parameters.

### 5.2 Results presentation and analysis

*5.2.1 Comparison with NSGA-II.* Based on the model formulation in Section 3 and the algorithm design in Section 4, this study compared the solution to the aforementioned real-world problem using both the NSGA-II and INSGA-II algorithms. The transportation data in this study, listed in Table 2, Table 3 and Table 6, are sourced from the operational data of a specific OEM. The model and its components, presented in Tables 5 and 6, were derived from the existing literature. The designed algorithm was implemented in MATLAB 2021b and executed on a personal computer equipped with an Intel Core-i7-1260P 2.10 GHz processor and 16 GB of RAM. Simulations were used to acquire the distribution of the Pareto-optimal solution sets and a hypervolume (HV) comparison of the traditional and enhanced NSGA-II algorithms for solving multi-objective Pareto-optimal solutions (Figure 6 and Figure 7, respectively). The hypervolume comparison shows that the HV value of the INSGA-II algorithm was approximately 10% higher than that of the conventional NSGA-II algorithm. Furthermore, in the Pareto-optimal solution set comparison, the enhanced NSGA-II exhibited a more pronounced layered Pareto front, which indicates different emphases on the objectives and variations in vehicle allocation, route selection and time scheduling. Additionally, the enhanced NSGA-II algorithm outperformed the traditional algorithm in terms of cost, carbon emissions and time satisfaction for most of the solutions obtained.

This study focused on three objective functions in path planning: minimizing costs and carbon emissions and maximizing time satisfaction. The optimal solutions for each objective

**Table 5.** Other fixed parameter values

| Parameter | Values |
|---|---|
| $\mu_1$ | 35 |
| $\mu_2$ | 65 |
| $\gamma$ | 1.471 |
| $\tau$ | 2.925 |

**Table 6.** Algorithm-related parameters

| Parameters | Values |
|---|---|
| NP | 150 |
| maxgen | 200 |
| $P_c$ | 0.8 |
| $P_m$ | 0.1 |

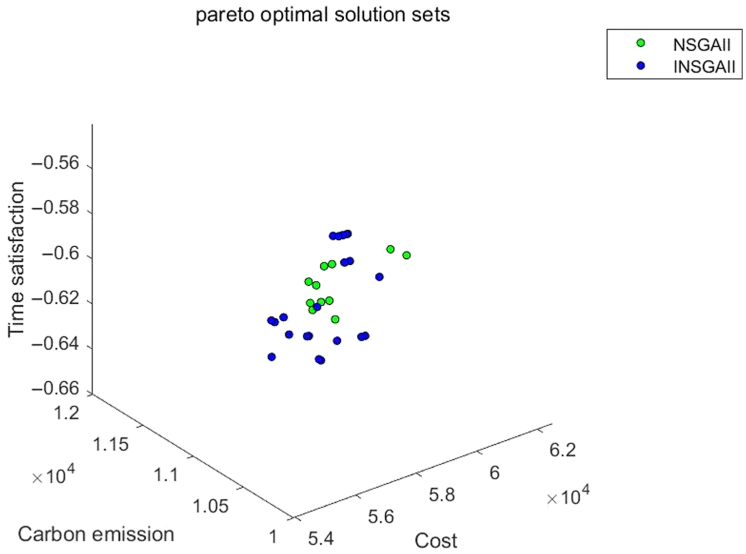


**Figure 6.** Comparison of the distribution of Pareto-optimal solution sets

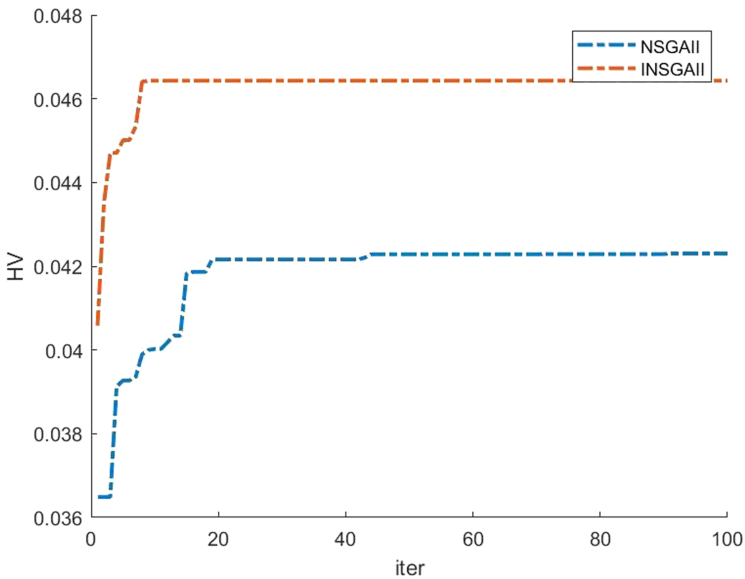


**Figure 7.** HV comparison

are listed in Table 7. Time satisfaction was quantified based on the delivery rate within the optimal time window. In reality, the main requirement for inbound logistics at the OEM is cargo arrival within an acceptable time window. Therefore, although the satisfaction level obtained in this study may be low, the cargo from all 30 suppliers arrived within an acceptable time window. Table 6 indicates a certain degree of directional consistency between carbon emissions and the

**Table 7.** Different focus on optimal solution results

| Programs | Objective 1 Cost (RMB) | Objective 2 Carbon emission (kg) | Objective 3 Time satisfaction (%) |
|---|---|---|---|
| 1: Cost priority | 53913 | 10911 | 60 |
| 2: Carbon emissions priority | 55975 | 10092 | 62 |
| 3: Time satisfaction priority | 56367 | 10200 | 64 |

time satisfaction objectives. During the model solving process, the solutions that prioritized Objectives 2 and 3 exhibited relatively small differences in carbon emissions and time satisfaction. The optimal solutions of Objectives 2 and 3 differed significantly from those of Objective 1. Cost reduction was associated with an increase in carbon emissions and a decrease in time satisfaction. However, a reduction in carbon emissions and improvement in time satisfaction results in higher costs. The enhanced NSGA-II algorithm was evaluated on three different objective paths: cost, carbon emission and time satisfaction, as shown in Figure 8, Figure 9 and Figure 10.

Currently, most companies prioritize cost as the primary or sole objective in path optimization for real-world operations. Substantial variation exists in carbon tax rates across countries. The highest rate is 1.207 RMB per kg of $CO_2$, whereas the global average is approximately 0.321 RMB per kg. For the scenario examined in this study, the optimal solution under the cost-minimization objective yielded a total carbon emission of 11311 kg, which is 819 kg higher than the emission under the low-carbon objective. The cost-saving objective with the high emission rate, however, reduced cost by 2062 RMB as compared to the low-carbon objective. At the highest carbon tax rate, the additional tax incurred for additional emissions would be 989 RMB. Thus, selecting the cost-optimal solution yielded an overall net cost reduction of 1264 RMB. Therefore, from a narrow economic perspective, companies tend

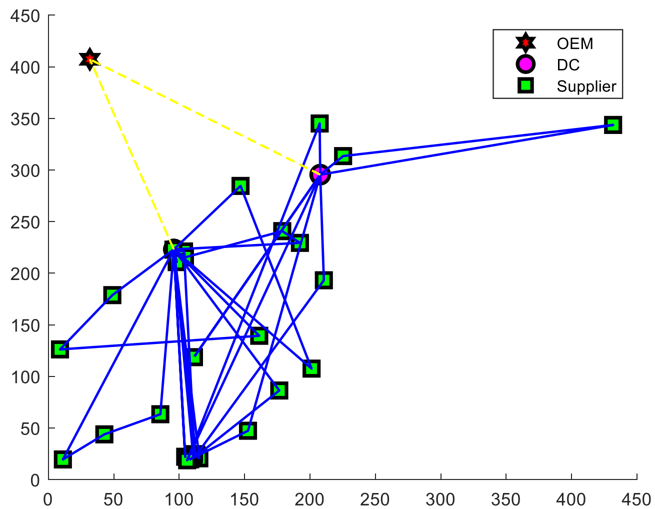


**Figure 8.** Cost-driven optimal paths in car inbound logistics

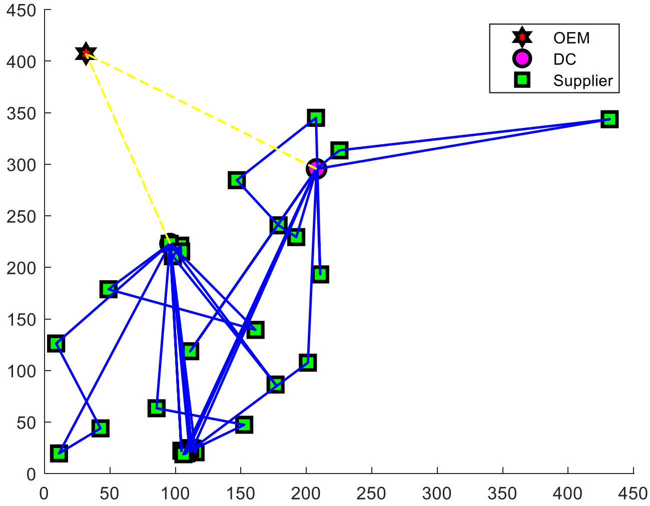


**Figure 9.** Carbon emission-driven optimal paths

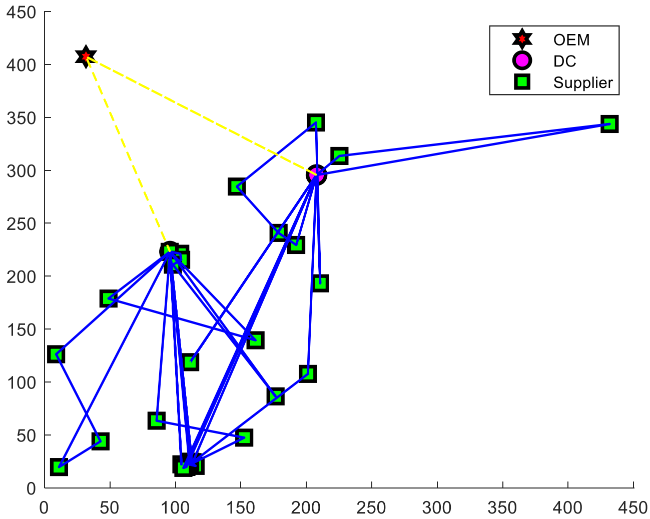


**Figure 10.** Time satisfaction-driven optimal paths

to treat cost as a decisive factor; carbon emissions and time satisfaction are placed as secondary considerations.

*5.2.2 Comparison with other algorithms.* Next, based on historical data from the municipal, provincial and interprovincial operations of an OEM, a benchmark case set with varying complexities was constructed to conduct comparative experimental analyses between the INSGA2 and other metaheuristic algorithms. The case set comprises nine representative instances, which are categorized into three groups based on the number of supplier nodes, with three instances in each group, as follows.

(1) Small-scale cases: The number of supply points is approximately 10. Such problems typically correspond to a municipal-level supply of commodity-grade components or the supply of advanced or large component assemblies (such as engines and seats) located close to the OEM. Their solution space is relatively limited and is often used to validate the basic performance of the algorithm in terms of solution quality and convergence rate.

(2) Medium-scale cases: The number of supply points ranges from several tens. These problems simulate daily delivery tasks for core components within a province or regional consolidation and transportation and serve as typical scenarios for testing the practicality of the algorithm. These scenarios retain sufficient complexity to challenge the search capability of the algorithm while maintaining the computation time within an acceptable range, thus rendering them suitable for in-depth parameter sensitivity analysis and performance comparisons between algorithms.

(3) Large-scale cases: The number of supply points exceeds 100. These problems correspond to the planning of large-scale interprovincial trunk line transportation networks. Their solution space grows exponentially and can be used to examine the computational efficiency and global exploration capabilities of the algorithm.

By encompassing data of varying scales, this study aims to investigate the comprehensive performance of the proposed algorithm in terms of solution quality and convergence stability in comparison to other multi-objective optimization algorithms when addressing the 2E-VRPSPDTW problem.

Several mainstream heuristic algorithms for multi-objective optimization from the current literature, MOEA/D (García-Nájera *et al.*, 2015), SPEA2 (Lv and Shen, 2023) and Pareto-TS2 (Ru, 2024), were used to serve as comparative algorithms to validate the effectiveness of INSGA2. The inverted generational distance (IGD) and hypervolume indicator (HV) were employed as performance metrics for the algorithm evaluation. For each test instance, the maximum number of function evaluations (NFEs) was set to 1,000 for small- and medium-scale cases and 2,000 for large-scale cases. All algorithms were implemented in MATLAB 2021b and executed on a PC equipped with an Intel Core-i7-1260P 2.10 GHz processor and 16 GB of RAM.

For the three comparative algorithms, MOEA/D, SPEA2 and Pareto-TS2, parameters such as population size, crossover probability and mutation probability were set to be consistent with those of INSGA2. The other key parameters were configured according to their respective original reference papers. The parameter settings for each algorithm were determined based on the literature and experimental fine-tuning to ensure convergence and diversity in solving the VRP problem. The specific parameter values are listed in Table 8.

Table 9 shows that INSGA2 achieved the best values across most objective functions. Specifically, in small-scale cases, the average values of INSGA2 for the three objective functions were 7.02E+3, 1.28E+3 and 7.25E+1, which were significantly lower than those of the other algorithms. Furthermore, its standard deviations were generally smaller, which indicates that INSGA2 not only provided higher solution accuracy but also exhibited better

**Table 8.** Key parameter values for the comparative algorithms

| Algorithm | Parameter symbol | Description | Value |
|---|---|---|---|
| MOEA/D | T | Neighborhood size | 20 |
| SPEA2 | A | Archive size | 100 |
| Pareto-TS | L | Tabu list size | 10 |
| | K | Neighborhood size | 50 |

**Table 9.** Comparison of solution quality

| Scale | Priority | INSGA2 | MOEA/D | SPEA2 | Pareto-TS |
|---|---|---|---|---|---|
| Small | Objective 1 (RMB) | 7.02E+3 (1.25E+2) | 7.85E+3 (1.68E+2) | 8.38E+3 (2.15E+2) | 8.52E+3 (2.46E+2) |
| | Objective 2 (kg) | 1.28E+3 (2.15E+1) | 1.52E+3 (3.24E+1) | 1.81E+3 (4.52E+1) | 1.83E+3 (4.68E+1) |
| | Objective 3 (%) | 7.25E+1 (1.52E+0) | 6.92E+1 (1.84E+0) | 6.68E+1 (2.15E+0) | 6.65E+1 (2.46E+0) |
| Medium | Objective 1 (RMB) | 5.41E+4 (3.25E+2) | 5.65E+4 (4.68E+2) | 5.91E+4 (5.25E+2) | 5.95E+4 (6.32E+2) |
| | Objective 2 (kg) | 1.01E+4 (1.52E+2) | 1.12E+4 (2.15E+2) | 1.24E+4 (3.25E+2) | 1.28E+4 (3.68E+2) |
| | Objective 3 (%) | 6.38E+1 (1.25E+0) | 6.15E+1 (1.68E+0) | 6.05E+1 (2.15E+0) | 5.98E+1 (2.46E+0) |
| Large | Objective 1 (RMB) | 1.42E+5 (1.52E+3) | 1.41E+5 (1.84E+3) | 1.57E+5 (2.15E+3) | 1.68E+5 (2.46E+3) |
| | Objective 2 (kg) | 3.02E+4 (3.25E+2) | 2.98E+4 (3.68E+2) | 3.42E+4 (4.52E+2) | 3.58E+4 (5.25E+2) |
| | Objective 3 (%) | 6.05E+1 (1.52E+0) | 6.12E+1 (1.84E+0) | 5.82E+1 (2.15E+0) | 5.65E+1 (2.46E+0) |

stability when solving small-scale problems. INSGA2 yielded the best results across all three objectives for the medium-scale cases, with values of 5.41E+4, 1.01E+4 and 6.38E+1, thus further validating the superiority of the algorithm in complex scenarios. Notably, in large-scale cases, while the performance of INSGA2 on Objectives 1 and 2 was similar to that of MOEA/D, INSGA2 maintained a slight lead on Objective 3. Overall, INSGA2 demonstrated a clear advantage in terms of solution quality. These results indicate that the INSGA2 algorithm can effectively adapt to problems of different scales and provide higher-quality Pareto-optimal solutions in multi-objective optimization.

An analysis of the IGD metric demonstrated the superior convergence performance of INSGA2 across all cases. Table 10, which lists the results for the small-scale scenario, yields IGD values for the INSGA2 of 8.52E−2, 7.92E−2 and 8.32E−2 for runs 1–3, respectively, which are lower than those of the other algorithms. The small standard deviations indicate good repeatability and stability across multiple runs. In medium-scale instances 4–6, the INSGA2 yielded the lowest IGD values as well, 1.25E−1, 1.18E−1 and 1.22E−1, respectively. These results further highlight the good convergence capability of INSGA2 for medium-complexity problems. In large-scale instances 7–9, although MOEA/D yielded slightly lower IGD values for some instances, INSGA2 remained superior in most cases and

**Table 10.** IGD comparison

| Scale | Ins | INSGA2 | MOEA/D | SPEA2 | Pareto-TS |
|---|---|---|---|---|---|
| Small | 1 | 8.52E−2 (1.21E−2) | 9.23E−2 (1.51E−2) | 1.05E−1 (1.81E−2) | 1.18E−1 (2.21E−2) |
| | 2 | 7.92E−2 (1.01E−2) | 8.82E−2 (1.31E−2) | 9.85E−2 (1.61E−2) | 1.12E−1 (2.01E−2) |
| | 3 | 8.32E−2 (1.11E−2) | 9.05E−2 (1.41E−2) | 1.02E−1 (1.71E−2) | 1.15E−1 (2.11E−2) |
| Medium | 4 | 1.25E−1 (1.81E−2) | 1.42E−1 (2.21E−2) | 1.68E−1 (2.51E−2) | 1.95E−1 (3.01E−2) |
| | 5 | 1.18E−1 (1.61E−2) | 1.35E−1 (2.01E−2) | 1.58E−1 (2.31E−2) | 1.82E−1 (2.81E−2) |
| | 6 | 1.22E−1 (1.71E−2) | 1.40E−1 (2.11E−2) | 1.62E−1 (2.41E−2) | 1.88E−1 (2.91E−2) |
| Large | 7 | 1.85E−1 (2.51E−2) | 1.78E−1 (2.31E−2) | 2.25E−1 (3.21E−2) | 2.85E−1 (4.01E−2) |
| | 8 | 1.82E−1 (2.61E−2) | 1.85E−1 (2.41E−2) | 2.35E−1 (3.41E−2) | 2.95E−1 (4.21E−2) |
| | 9 | 1.88E−1 (2.51E−2) | 1.82E−1 (2.31E−2) | 2.28E−1 (3.31E−2) | 2.90E−1 (4.11E−2) |

showed superior performance in instances 7 and 9. Overall, the superior IGD results of INSGA2 indicate that its generated solution set can approximate the true Pareto front more closely than the other algorithms, thereby demonstrating strong global exploration and convergence capabilities.

INSGA2 also exhibited excellent comprehensive performance in terms of the HV metric. The HV value reflects the ability of the algorithm to balance the convergence and diversity of the solution set, with a higher HV value typically indicating a better overall performance. As shown in Table 11, in small-scale instances, the INSGA2 yielded higher HV values than the other algorithms, 5.12E−1, 5.98E−1 and 5.05E−1, respectively, and its standard deviations were small, thus indicating good robustness in maintaining the uniformity of the solution set distribution. The INSGA2 yielded the best HV values at the medium-scale as well, with values ranging between 4.76E−1 and 4.85E−1. In large-scale instances, although the HV values of INSGA2 were slightly lower than those of MOEA/D in some cases, the INSGA2 outperformed SPEA2 and Pareto TS overall, with the best performance observed for instance 8. These results indicate that the INSGA2 can effectively maintain the diversity and convergence of the solution set across different problem scales, thereby providing a more representative Pareto front.

In conclusion, the INSGA2 algorithm demonstrated superior comprehensive performance across the three key metrics of solution quality, IGD and HV, with particularly significant leads in small- and medium-scale cases. These results show that when solving the milk-run-based automotive inbound logistics optimization problem considering empty container flow, INSGA2 can not only converge quickly to high-quality solutions but also maintain good distribution characteristics within the solution set. Furthermore, the algorithm maintained relatively optimal performance in large-scale problems, indicating good scalability and robustness. Therefore, the INSGA2 algorithm holds high practical value and potential for wider application in addressing complex logistics optimization scenarios.

*5.2.3 Sensitivity analysis.* The impact of time penalty costs and carbon tax variations on the route optimization results was further analyzed by multiplying the minimum value of Objective 2 (carbon emissions) by a unit carbon tax rate to transform it into an objective function directly linked to carbon tax costs. Based on a medium-scale case, five scenarios for comparative analysis were established: baseline (BAS), high time penalty cost (HPC), low time penalty cost (LPC), high carbon tax (HCT) and low carbon tax (LCT) scenarios. The corresponding results for the total cost, carbon emissions and time satisfaction under each scenario are shown in Figure 11. The curves in the figure quantify the relative performance of INSGA2 compared to NSGA-II and represent the percentage difference between the original NSGA-II and the proposed INSGA2. For Objectives 1 and 2 (minimization objectives), a higher positive value indicates a more significant advantage of INSGA2 over NSGA-II; positive and large values indicate superiority. For Objective 3 (maximization objective), when

**Table 11.** HV comparison

| Scale | Ins | INSGA2 | MOEA/D | SPEA2 | Pareto-TS |
|---|---|---|---|---|---|
| Small | 1 | 5.12E−1 (1.52E−2) | 4.84E−1 (1.84E−2) | 4.52E−1 (2.15E−2) | 4.38E−1 (2.46E−2) |
| | 2 | 5.98E−1 (1.48E−2) | 4.69E−1 (1.79E−2) | 4.41E−1 (2.12E−2) | 4.26E−1 (2.43E−2) |
| | 3 | 5.05E−1 (1.55E−2) | 4.77E−1 (1.82E−2) | 4.48E−1 (2.18E−2) | 4.31E−1 (2.51E−2) |
| Medium | 4 | 4.85E−1 (1.65E−2) | 4.58E−1 (1.95E−2) | 4.23E−1 (2.25E−2) | 4.05E−1 (2.65E−2) |
| | 5 | 4.76E−1 (1.68E−2) | 4.46E−1 (1.98E−2) | 4.12E−1 (2.32E−2) | 3.94E−1 (2.72E−2) |
| | 6 | 4.81E−1 (1.62E−2) | 4.52E−1 (1.92E−2) | 4.18E−1 (2.28E−2) | 4.01E−1 (2.68E−2) |
| Large | 7 | 4.23E−1 (1.85E−2) | 4.35E−1 (2.05E−2) | 4.02E−1 (2.45E−2) | 3.85E−1 (2.85E−2) |
| | 8 | 4.28E−1 (1.82E−2) | 4.48E−1 (2.08E−2) | 4.15E−1 (2.42E−2) | 3.98E−1 (2.82E−2) |
| | 9 | 4.21E−1 (1.88E−2) | 4.32E−1 (2.12E−2) | 4.09E−1 (2.48E−2) | 3.92E−1 (2.88E−2) |

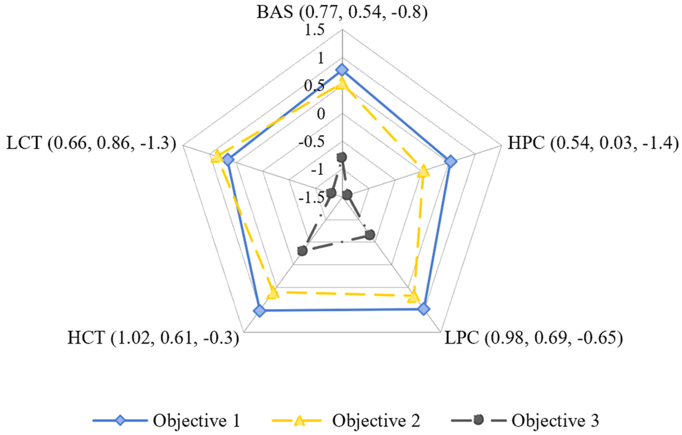


**Figure 11.** Parameter sensitivity analysis

the indicator value is negative, a larger absolute value indicates the superior service performance of INSGA2 compared with NSGA-II.

The results indicate that changes in time penalty costs and carbon taxes significantly affect all objective metrics of the system. Under the HPC scenario, to reduce penalties from delayed deliveries, the model prioritizes ensuring on-time delivery rates, potentially by increasing vehicle frequency or selecting faster but more expensive routes to improve service levels. Consequently, transportation costs and carbon emissions increase. Conversely, under the LPC scenario, the model can relax time constraints moderately to focus on optimizing cost and carbon emissions, thus performing better in terms of total cost and emissions, although time satisfaction may decrease slightly.

Regarding carbon tax, under the HCT scenario, the weight of carbon emission costs within the total cost increases significantly. To reduce carbon-related expenses, the model tends to favor more environmentally friendly transportation plans, such as optimizing vehicle loading rates, reducing empty mileage and prioritizing vehicles with higher energy efficiency, which leads to outstanding carbon emission metrics. However, this selection may incur increased operational costs due to route adjustments or vehicle redeployment. In contrast, under the LCT scenario, the pressure from carbon emission costs diminishes and the model leans more toward pursuing an operational economy, which may, in turn, yield an increase in carbon emissions.

In summary, as external policy and economic levers, time penalty costs and carbon taxes have a clear directional impact on the multi-objective coordination of the route optimization system. An increase in time penalty costs strengthens the constraint of service timeliness by pushing the system toward a "service-first" orientation. An increase in the carbon tax enhances environmental constraints and drives the system toward green transportation. To a certain extent, both factors create a trade-off with the cost objective, which reflects inherent conflicts and coordination potential among the economic, environmental and service dimensions.

## 6. Policy recommendations and managerial implications

### *6.1 Policy recommendations*

The low-carbon transition in the transportation sector has become a crucial component contributing toward the "dual carbon" goals. This study addresses the route optimization problem for milk-run-based automotive inbound logistics by constructing a multi-objective decision-making model that integrates cost, carbon emissions and time satisfaction. The effectiveness of the model and algorithm was validated through case studies. Based on the research findings, the following policy recommendations are proposed to promote the greening and intensification of automotive logistics:

(1) Refine Carbon Pricing and Green Logistics Incentive Policies. A differentiated carbon tax system should be established based on transportation characteristics of automotive inbound logistics. Carbon tax reductions or exemptions should be offered to logistics companies that adopt the milk-run system and implement simultaneous pickup and delivery operations. Enterprises that do not employ green transportation modes and have empty-run rates above industry benchmarks should be subject to a moderate increase in carbon tax rates.

(2) Promote Standardized Container Systems and Circular Sharing Mechanisms. Thus, policies should further promote the standardization and modularization of automotive parts logistics containers and support the establishment of an industry-level platform for sharing and coordinating empty containers.

(3) Encourage Cross-Enterprise Collaboration and Infrastructure Sharing. The government should promote the establishment of regional automotive logistics collaboration platforms through demonstration projects and financial support. These platforms would support multiple enterprises in sharing distribution centers, transportation vehicles and information systems to enhance network operational efficiency and achieve economies of scale in emission reduction.

(4) Establish a Dynamic Policy Adjustment Mechanism. A feedback mechanism for adjusting policies should be established based on carbon reduction outcomes and industry development trends. In line with advancements in new-energy logistics vehicle technology, vehicle subsidy policies should be adjusted in a timely manner to encourage logistics companies to gradually replace high-fuel-consumption vehicles and promote the application of electrified and intelligent vehicles in milk-run operations.

*6.2 Managerial implications*

This section provides actionable management guidance for logistics managers at OEMs and dispatchers at distribution centers and discusses challenges as well as mitigation strategies associated with implementing the proposed algorithm to facilitate the translation of theoretical findings into practice.

*6.2.1 Key insights.*

(1) Dynamic Optimization of Multi-objective Weights. The trade-off between cost, carbon emissions and time satisfaction must be dynamically adjusted in route planning under the milk-run system according to the corporate strategy and the external environment.

(2) Role of Container Status Tracking in Enhancing Routing Efficiency. OEM logistics managers and dispatchers should establish a full-process container status tracking system. During route planning, the sequence of pickups and empty container returns should be optimized based on each supplier's demand for empty containers and the supply of full containers to improve vehicle loading rates. Thus, an early warning mechanism for container turnover should be established. When a supplier's empty container inventory falls below the safety threshold, the milk-run frequency can be adjusted to prevent component supply disruptions due to empty container shortages or increased warehousing and transportation costs due to empty container stockpiling.

(3) Scenario-based Selection between Milk-run and Direct Shipping. Dispatchers should clearly define applicability boundaries for different transportation modes. When

suppliers are widely dispersed (e.g. across provinces) and their empty container demands vary significantly, a hybrid model of "regional milk-run and trunk-line direct shipping" can be adopted, which implements the milk-run to consolidate goods and empty containers within regions and then utilizes trunk-line shipping to the OEM to balance efficiency and cost.

(4) Management Response to Carbon Tax Sensitivity and Policy Anticipation. OEM logistics managers should establish a carbon tax trend anticipation mechanism to closely monitor changes in carbon pricing policies and proactively optimize the transportation network to mitigate future cost pressures from potential carbon tax increases. Simultaneously, collaborative operations among the OEM, suppliers and distribution centers should be promoted alongside optimization of loading/unloading processes, improvements in cargo consolidation efficiency and reductions in the vehicle idle time and empty runs caused by unsynchronized operations. These implementations enhance the incentive effect of carbon tax policies on the low-carbon transition.

*6.2.2 Implementation challenges.* The following challenges may be encountered when the proposed model is applied to practical operations. Mitigation measures are proposed for each of the challenges mentioned.

(1) Data Availability and Quality Challenges. Route optimization for the milk-run system relies on multidimensional data such as supplier schedules, cargo weight, container quantities and time windows. However, data gaps, outdated information and inaccuracies present common issues in practice and cause a disconnection between model optimization results and reality. Enterprises should therefore promote data standardization across supply chains and clarify data provision responsibilities through service agreements. Simultaneously, techniques such as data cleansing and outlier handling can be employed to improve data quality; appropriate safety buffers (e.g. time buffers) should be set to account for data uncertainty.

(2) Model and Algorithm Scalability. A significant growth in the number of suppliers is associated with an exponential growth in the computational complexity of the constructed mixed-integer programming model and improved NSGA-II algorithm. This complexity may reduce route planning efficiency and thus render meeting real-time scheduling demands difficult. Enterprises can adopt strategies such "hierarchical planning" or "zonal optimization," which divide a large-scale transportation network into several zones, perform independent milk-run route planning within each zone and consolidate interzonal cargo via trunk-line transportation. This approach reduces the problem size for a single optimization run.

(3) Integration with OEM transportation management systems (TMS). Existing OEM TMS are often designed for traditional transportation modes and lack support for features specific to this study, such as the milk-run system, simultaneous delivery and pickup and nestable empty-container handling. This limitation hinders seamless integration of the proposed optimization model with existing systems and thus limits the practical application of the theoretical findings. Enterprises can develop modular interface programs and design independent modules for the core functions of the model to connect with existing TMS, thereby avoiding the high cost of system reengineering. Furthermore, a phased integration strategy–pilot integration in a specific region – can be adopted to validate the model's effectiveness and system compatibility before a broader rollout.

(4) Sensitivity to Operational Uncertainty. Vehicle travel time, supplier loading/unloading duration and cargo demand quantities inherently contain a high level of

uncertainty. Thus, logistic service quality and carbon reduction outcomes may be negatively impacted as optimization paths based on deterministic data are rendered ineffective. Enterprises must therefore establish real-time dynamic scheduling mechanisms by utilizing technologies such as GPS tracking and IoT monitoring to obtain real-time data. The use of such mechanisms can cause the model to reoptimize and adjust transportation routes and vehicle dispatch plans when deviations from the plan exceed a predefined threshold. In addition, buffers with reasonable time and capacity should be set. For example, reserving 10%–15% time redundancy in vehicle scheduling and maintaining 20% flexible storage capacity at distribution centers can help cope with sudden demand changes and delay risks.

## 7. Conclusion

This study examines the route optimization problem of automotive inbound logistics under “dual carbon” constraints, specifically addressing the “simultaneous pickup and delivery” operational characteristic of the milk-run mode. A multi-objective optimization model integrating economic, environmental and service timeliness objectives was constructed, and an INSGA-II algorithm was designed. The following conclusions were drawn from the theoretical analysis and case validation.

(1) The constructed multi-objective mixed-integer programming model, which incorporates nestable empty container constraints, multi-phase operations and carbon emission costs, accurately captures the network structure and operational characteristics of automotive milk-run logistics. Thus providing a modeling framework for optimizing transportation costs, carbon emissions and time satisfaction simultaneously.

(2) By designing targeted encoding, crossover and mutation strategies, the search efficiency and convergence performance of the algorithm for solving the complex 2E-VRPSPDTW were enhanced. Simulation experiments and comparative analyses with multiple algorithms demonstrated that INSGA-II outperforms advanced meta-heuristic methods such as MOEA/D, SPEA2 and Pareto-TS in terms of solution quality, convergence (IGD metric) and distribution (HV metric). Its performance was particularly outstanding in small- and medium-scale cases, thus validating the effectiveness and superiority of the algorithm for handling this problem.

(3) The model and algorithm proposed in this study can provide decision-makers with differentiated route planning solutions. This study quantitatively revealed the inherent conflicts among the three optimization objectives. For OEM logistics planners, the pursuit of environmental benefits or service improvements is not cost-free. Considering the medium-scale case as an example, achieving a 7.5% reduction in carbon emissions directly leads to a 3.8% increase in the total logistics cost. A simultaneous improvement in the time satisfaction by 6.7% from the industry benchmark resulted in an additional 4.5% cost increment. These findings indicate that, under the current technological and managerial framework, no single solution can simultaneously achieve absolute optimality in cost, carbon emissions and time satisfaction. Thus, the planner must dynamically adjustment objective weights based on corporate strategic priorities, such as cost control, carbon reduction or lean production requirements, which involves a trade-off between marginal cost and benefit.

(4) Time penalty costs and carbon taxes have a clear directional effect on the optimization outcome. The study confirms that increasing time penalty costs strengthens the service timeliness constraint by pushing solutions toward a “service-first” adjustment. A high carbon tax reinforces the “green transportation” orientation and incentivizes the

selection of low-carbon routes. This finding provides an important basis for enterprises to respond to changes in external policies and adjust their operational strategies dynamically.

This study has certain limitations that warrant further exploration in future work. The model was based on several deterministic assumptions and did not fully account for the dynamics and uncertainties in actual operations, such as traffic congestion and demand fluctuations. Future research could introduce stochastic programming or robust optimization methods to enhance the resilience of the model to disturbances. Moreover, the computational efficiency of this algorithm can be improved for large-scale problems. Exploring intelligent optimization frameworks based on reinforcement learning or distributed computing presents a promising direction for future work.

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